



A more efficient reformulation of complex SDP as real SDP

Jie Wang¹

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Abstract

This note proposes a new reformulation of complex semidefinite programs (SDP) as real SDPs. As an application, we present an economical reformulation of complex SDP relaxations of complex polynomial optimization problems as real SDPs and derive some further reductions by exploiting inner structure of the complex SDP relaxations. Various numerical examples demonstrate that our new reformulation runs significantly faster than the usual popular reformulation.

Keywords complex semidefinite programming · complex polynomial optimization · semidefinite programming · the complex moment-HSOS hierarchy · quantum information

Mathematics Subject Classification 90C22 · 90C23

1 Introduction

Complex semidefinite programs (SDP) arise from a diverse set of areas, such as combinatorial optimization [11], optimal power flow [12, 14], quantum information theory [2, 4, 19], and signal processing [13, 16]. In particular, they appear as convex relaxations of complex polynomial optimization problems (CPOP), giving rise to the complex moment-Hermitian-sum-of-squares (moment-HSOS) hierarchy [12, 17, 18].

✉ Jie Wang
wangjie212@amss.ac.cn
<https://wangjie212.github.io/jiewang/>

¹ Academy of Mathematics and Systems Science, Chinese Academy of Sciences, Beijing, China

However, most modern SDP solvers deal with only real SDPs¹. In order to handle complex SDPs via real SDP solvers, it is mandatory to reformulate complex SDPs as equivalent real SDPs. A popular way² to do so is to use the equivalent condition

$$H \succeq 0 \iff Y = \begin{bmatrix} H_R & -H_I \\ H_I & H_R \end{bmatrix} \succeq 0 \quad (1.1)$$

for a Hermitian matrix variable $H = H_R + H_I \mathbf{i} \in \mathbb{C}^{n \times n}$ with H_R and H_I being its real and imaginary parts respectively. Note that the right-hand-side constraint in (1.1) entails certain structure and to feed it to an SDP solver, we need to impose extra affine constraints to the positive semidefinite (PSD) constraint $Y \succeq 0$:

$$Y_{i,j} = Y_{i+n,j+n}, \quad Y_{i,j+n} + Y_{j,i+n} = 0, \quad i = 1, \dots, n, \quad j = i, \dots, n. \quad (1.2)$$

This conversion is quite simple but could be inefficient when n is large. In this note, inspired by Lagrange duality, we propose a new reformulation of complex SDPs as real SDPs. The benefit of this new reformulation is that it contains no extra affine constraints while keeping the same matrix size and hence has lower complexity. In the same manner, we can obtain a new reformulation of complex SDP relaxations of CPOPs as real SDPs. Moreover, by exploiting inner structure of the complex SDP relaxations, we are able to remove a bunch of redundant affine constraints, which leads to an even more economical real reformulation of the complex SDP relaxations. Various numerical experiments (including the complex matrix completion problem, complex SDP relaxations for randomly generated CPOPs, and the alternating current optimal power flow (AC-OPF) problem) confirm our theoretical expectation and demonstrate that the new reformulation is substantially more efficient than the usual popular one. Finally, numerical experiments also indicate that our implementation of the new reformulation with MOSEK [1] runs much faster than the implementation of the original complex formulation with Hypatia [10], probably because MOSEK is commercial and the implementations of SDP algorithms based on real numbers are more mature and robust.

Notation The symbol $\mathbb{N}$ denotes the set of nonnegative integers. For $n \in \mathbb{N} \setminus \{0\}$, let $[n] := \{1, 2, \dots, n\}$. We write $|A|$ for the cardinality of a set A . Let $\mathbf{i}$ be the imaginary unit, satisfying $\mathbf{i}^2 = -1$. For $d \in \mathbb{N}$, let $\mathbb{N}_d^n := \{(\alpha_i)_i \in \mathbb{N}^n \mid \sum_{i=1}^n \alpha_i \leq d\}$ and let $\omega_{n,d} := \binom{n+d}{d}$ be the cardinality of $\mathbb{N}_d^n$. For $\alpha = (\alpha_i)_i \in \mathbb{N}_d^n$ and an n -tuple of variables $\mathbf{z} = \{z_1, \dots, z_n\}$, let $\mathbf{z}^\alpha := z_1^{\alpha_1} \cdots z_n^{\alpha_n}$. For a complex number a , $\bar{a}$ (resp. $\mathcal{R}(a)$, $\mathcal{I}(a)$) denotes the conjugate (resp. real part, imaginary part) of a , and for a complex vector v , v^H denotes the conjugate transpose of v . For a positive integer n , the set of $n \times n$ symmetric (resp. Hermitian) matrices is denoted by $\mathbf{S}^n$ (resp. $\mathbf{H}^n$). We use $A \succeq 0$ to indicate that the matrix A is PSD. For $A, B \in \mathbb{C}^{n \times n}$, we denote by

¹ As far as the author knows, SeDuMi [15], Sdolab [9], and Hypatia [8] are the only solvers that can handle complex SDPs directly.

² See for instance the online modeling cookbook of the commercial SDP solver MOSEK: <https://docs.mosek.com/modeling-cookbook/sdo.html>.

$\langle A, B \rangle$ the trace inner-product, defined by $\langle A, B \rangle = \text{Tr}(A^H B)$, where A^H stands for the conjugate transpose of A . For $A \in \mathbb{R}^{n \times n}$, $A^\top$ stands for the transpose of A .

We endow $\mathbb{C}^m$ (viewed as a $\mathbb{R}$ -vector space) with the scalar product $\langle \cdot, \cdot \rangle_{\mathbb{R}}$ defined by

$$\langle u, v \rangle_{\mathbb{R}} = \mathcal{R}(u^H v) = \mathcal{R}(u)^\top \mathcal{R}(v) + \mathcal{I}(u)^\top \mathcal{I}(v), \quad u, v \in \mathbb{C}^m,$$

which will be used in forming the dual complex SDP problem (to obtain a real objective). For $u, v \in \mathbb{R}^m$, $\langle u, v \rangle := u^\top v$, where $u^\top$ stands for the transpose of u . We associate each $A \in \mathbb{C}^{n \times n}$ with a Hermitian matrix $\mathcal{H}(A) := \frac{1}{2}(A + A^H)$. One can check that $\mathcal{R}(\langle A, H \rangle) = \langle \mathcal{H}(A), H \rangle$ for any $H \in \mathbb{H}^n$.

2 The real reformulations of complex SDPs

Given a tuple of complex matrices $A_1, \dots, A_m \in \mathbb{C}^{n \times n}$, we define a $\mathbb{R}$ -linear operator $\mathcal{A}: \mathbb{H}^n \rightarrow \mathbb{C}^m$ by

$$\mathcal{A}(H) := (\langle A_i, H \rangle)_{i=1}^m \in \mathbb{C}^m, \quad \forall H \in \mathbb{H}^n. \quad (2.1)$$

Let us consider the following complex SDP:

$$\begin{cases} \sup_{H \in \mathbb{H}^n} \langle C, H \rangle \\ \text{s.t. } \mathcal{A}(H) = b, \\ H \succeq 0, \end{cases} \quad (\text{PSDP-}\mathbb{C})$$

where $C \in \mathbb{H}^n$, $b \in \mathbb{C}^m$. In order to convert (PSDP- $\mathbb{C}$) to a real SDP, we define two real linear operators $\mathcal{A}_R, \mathcal{A}_I: \mathbb{R}^{n \times n} \rightarrow \mathbb{R}^m$ associated with $\mathcal{A}$ by

$$\mathcal{A}_R(S) := (\langle \mathcal{R}(A_i), S \rangle)_{i=1}^m \in \mathbb{R}^m, \quad \forall S \in \mathbb{R}^{n \times n} \quad (2.2)$$

and

$$\mathcal{A}_I(S) := (\langle \mathcal{I}(A_i), S \rangle)_{i=1}^m \in \mathbb{R}^m, \quad \forall S \in \mathbb{R}^{n \times n}, \quad (2.3)$$

respectively. Moreover, assume $H = H_R + H_I \mathbf{i}$, $C = C_R + C_I \mathbf{i}$, $b = b_R + b_I \mathbf{i}$ with $H_R, H_I, C_R, C_I \in \mathbb{R}^{n \times n}$, $b_R, b_I \in \mathbb{R}^m$. We can now convert (PSDP- $\mathbb{C}$) to a real SDP:

$$\begin{cases} \sup_{Y \in \mathbb{S}^{2n}} \langle C_R, H_R \rangle + \langle C_I, H_I \rangle \\ \text{s.t. } \mathcal{A}_R(H_R) + \mathcal{A}_I(H_I) = b_R, \\ \mathcal{A}_R(H_I) - \mathcal{A}_I(H_R) = b_I, \\ Y = \begin{bmatrix} H_R & -H_I \\ H_I & H_R \end{bmatrix} \succeq 0. \end{cases} \quad (\text{PSDP-}\mathbb{R})$$

As mentioned in the introduction, to feed the PSD constraint in (PSDP- $\mathbb{R}$) to an SDP solver, we need to include also the extra $n(n+1)$ affine constraints listed in (1.2),

which could be inefficient in practice. Below we show that by taking a dual point of view, we can actually get rid of this issue.

Before formulating the dual problem of (PSDP-C), we explicitly give the adjoint operator of $\mathcal{A}$.

Lemma 2.1 *The adjoint operator $\mathcal{A}^*$ of $\mathcal{A}$ satisfies $\mathcal{A}^*(y) = \mathcal{H}(\sum_{i=1}^m y_i A_i)$ for $y \in \mathbb{C}^m$.*

Proof For any $H \in \mathbb{H}^n$, we have

$$\begin{aligned} \langle \mathcal{A}^*(y), H \rangle &= \langle y, \mathcal{A}(H) \rangle_{\mathbb{R}} = \mathcal{R} \left(\sum_{i=1}^m \bar{y}_i \langle A_i, H \rangle \right) \\ &= \mathcal{R} \left(\left\langle \sum_{i=1}^m y_i A_i, H \right\rangle \right) = \left\langle \mathcal{H} \left(\sum_{i=1}^m y_i A_i \right), H \right\rangle, \end{aligned}$$

which yields $\mathcal{A}^*(y) = \mathcal{H}(\sum_{i=1}^m y_i A_i)$. $\square$

Then by the convex duality theory, the dual problem of (PSDP-C) reads as

$$\begin{cases} \inf_{y \in \mathbb{C}^m} & \langle b, y \rangle_{\mathbb{R}} \\ \text{s.t.} & \mathcal{A}^*(y) \succeq C. \end{cases} \quad (\text{DSDP-C})$$

Assume $y = y_R + y_I \mathbf{i}$ with $y_R, y_I \in \mathbb{R}^m$. Using Lemma 2.1, we deduce that $\mathcal{A}^*(y) = U + V \mathbf{i}$ with

$$U := \frac{1}{2} \sum_{i=1}^m \mathcal{R}(y_i) \mathcal{R}(A_i + A_i^{\top}) - \mathcal{I}(y_i) \mathcal{I}(A_i + A_i^{\top})$$

and

$$V := \frac{1}{2} \sum_{i=1}^m \mathcal{I}(y_i) \mathcal{R}(A_i - A_i^{\top}) + \mathcal{R}(y_i) \mathcal{I}(A_i - A_i^{\top}).$$

Thus, we can convert (DSDP-C) to a real SDP by using the equivalent condition (1.1):

$$\begin{cases} \inf_{y_R, y_I \in \mathbb{R}^m} & b_R^{\top} y_R + b_I^{\top} y_I \\ \text{s.t.} & \begin{bmatrix} U - C_R & -V + C_I \\ V - C_I & U - C_R \end{bmatrix} \succeq 0. \end{cases} \quad (\text{DSDP-R})$$

Let $X = \begin{bmatrix} X_1 & X_3^{\top} \\ X_3 & X_2 \end{bmatrix} \in \mathbb{S}^{2n}$ be the PSD variable of (DSDP-R) with $X_1, X_2, X_3 \in \mathbb{R}^{n \times n}$. Then the Lagrangian associated with (DSDP-R) is given by

$$\begin{aligned}
& L(X, y_R, y_I) \\
&= b_R^T y_R + b_I^T y_I - \left\langle \begin{bmatrix} X_1 & X_3^T \\ X_3 & X_2 \end{bmatrix}, \begin{bmatrix} U - C_R - V + C_I \\ V - C_I & U - C_R \end{bmatrix} \right\rangle \\
&= b_R^T y_R + b_I^T y_I - \langle X_1 + X_2, U - C_R \rangle - \langle X_3 - X_3^T, V - C_I \rangle \\
&= b_R^T y_R + b_I^T y_I + \langle C_R, X_1 + X_2 \rangle + \langle C_I, X_3 - X_3^T \rangle \\
&\quad - \sum_{i=1}^m \langle \mathcal{R}(A_i), X_1 + X_2 \rangle \mathcal{R}(y_i) + \sum_{i=1}^m \langle \mathcal{I}(A_i), X_1 + X_2 \rangle \mathcal{I}(y_i) \\
&\quad - \sum_{i=1}^m \langle \mathcal{R}(A_i), X_3 - X_3^T \rangle \mathcal{I}(y_i) - \sum_{i=1}^m \langle \mathcal{I}(A_i), X_3 - X_3^T \rangle \mathcal{R}(y_i) \\
&= \langle C_R, X_1 + X_2 \rangle + \langle C_I, X_3 - X_3^T \rangle + \langle b_R - \mathcal{A}_R(X_1 + X_2) - \mathcal{A}_I(X_3 - X_3^T), y_R \rangle \\
&\quad + \langle b_I - \mathcal{A}_R(X_3 - X_3^T) + \mathcal{A}_I(X_1 + X_2), y_I \rangle.
\end{aligned}$$

Therefore, the dual problem of (PSDP- $\mathbb{R}$) can be written down as

$$\begin{cases} \sup_{X \in \mathbb{S}^{2n}} & \langle C_R, X_1 + X_2 \rangle + \langle C_I, X_3 - X_3^T \rangle \\ \text{s.t.} & \mathcal{A}_R(X_1 + X_2) + \mathcal{A}_I(X_3 - X_3^T) = b_R, \\ & \mathcal{A}_R(X_3 - X_3^T) - \mathcal{A}_I(X_1 + X_2) = b_I, \\ & X = \begin{bmatrix} X_1 & X_3^T \\ X_3 & X_2 \end{bmatrix} \succeq 0. \end{cases} \quad (\text{PSDP-}\mathbb{R}')$$

Theorem 2.2 (PSDP- $\mathbb{R}'$) is equivalent to (PSDP- $\mathbb{R}$) (in the sense that they share the same optimum). As a result, (PSDP- $\mathbb{R}'$) is equivalent to (PSDP- $\mathbb{C}$). In addition, if $X^* = \begin{bmatrix} X_1^* & (X_3^*)^T \\ X_3^* & X_2^* \end{bmatrix}$ is an optimal solution to (PSDP- $\mathbb{R}'$), then $H^* = (X_1^* + X_2^*) + (X_3^* - (X_3^*)^T)\mathbf{i}$ is an optimal solution to (PSDP- $\mathbb{C}$).

Proof Let us denote the optima of (PSDP- $\mathbb{R}$) and (PSDP- $\mathbb{R}'$) by v and v' , respectively. Suppose that $Y = \begin{bmatrix} H_R & -H_I \\ H_I & H_R \end{bmatrix}$ is a feasible solution to (PSDP- $\mathbb{R}$). Then one can easily check that $X := \frac{1}{2}Y$ is a feasible solution to (PSDP- $\mathbb{R}'$). Moreover, we have $\langle C_R, X_1 + X_2 \rangle + \langle C_I, X_3 - X_3^T \rangle = \langle C_R, H_R \rangle + \langle C_I, H_I \rangle$ and it follows $v \leq v'$. On the other hand, suppose $X = \begin{bmatrix} X_1 & X_3^T \\ X_3 & X_2 \end{bmatrix}$ is a feasible solution to (PSDP- $\mathbb{R}'$). Since PSDness is preserved under basis transformation, we have

$$\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}^{-1} X \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} X_2 & -X_3 \\ -X_3^T & X_1 \end{bmatrix} \succeq 0,$$

and since PSDness is preserved under addition, we further have

$$Y = \begin{bmatrix} H_R & -H_I \\ H_I & H_R \end{bmatrix} := \begin{bmatrix} X_1 + X_2 & X_3^T - X_3 \\ X_3 - X_3^T & X_1 + X_2 \end{bmatrix} \succeq 0.$$

One can easily see that Y is a feasible solution to (PSDP- $\mathbb{R}$) and in addition, it holds $\langle C_R, H_R \rangle + \langle C_I, H_I \rangle = \langle C_R, X_1 + X_2 \rangle + \langle C_I, X_3 - X_3^T \rangle$. Thus $v \geq v'$, which proves the equivalence. The latter statement of the theorem is clear from the above arguments. $\square$

In contrast to (PSDP- $\mathbb{R}$), the PSD constraint in (PSDP- $\mathbb{R}'$) does not entails any special structure, and thus no extra affine constraint is required to describe it. This is why the conversion (PSDP- $\mathbb{R}'$) is more appealing than (PSDP- $\mathbb{R}$) from the computational perspective.

Remark 2.3 A similar reformulation to (PSDP- $\mathbb{R}'$) but for a restricted class of complex SDP relaxations of multiple-input multiple-output detection has appeared in [13].

3 Application to complex SDP relaxations for CPOPs

In this section, we apply the reformulation (PSDP- $\mathbb{R}'$) to complex SDP relaxations arising from the complex moment-HSOS hierarchy for CPOPs. A CPOP is given by

$$\begin{cases} \inf_{\mathbf{z} \in \mathbb{C}^s} f(\mathbf{z}, \bar{\mathbf{z}}) = \sum_{\beta, \gamma} b_{\beta, \gamma} \mathbf{z}^\beta \bar{\mathbf{z}}^\gamma \\ \text{s.t. } g_i(\mathbf{z}, \bar{\mathbf{z}}) = \sum_{\beta, \gamma} g_{\beta, \gamma}^i \mathbf{z}^\beta \bar{\mathbf{z}}^\gamma \geq 0, \quad i \in [t], \end{cases} \quad (\text{CPOP})$$

where $\bar{\mathbf{z}} := (\bar{z}_1, \dots, \bar{z}_s)$ stands for the conjugate of complex variables $\mathbf{z} := (z_1, \dots, z_s)$. The functions $f, g_1, \dots, g_t$ are real-valued polynomials in $\mathbf{z}$ and $\bar{\mathbf{z}}$ (their coefficients satisfy $b_{\beta, \gamma} = \overline{b_{\gamma, \beta}}$, $g_{\beta, \gamma}^i = \overline{g_{\gamma, \beta}^i}$). The *support* of f is defined by $\text{supp}(f) := \{(\beta, \gamma) \mid b_{\beta, \gamma} \neq 0\}$. For $i \in [t]$, $\text{supp}(g_i)$ is defined in the same way.

Fix a $d \in \mathbb{N}$. Let $y = (y_{\beta, \gamma})_{(\beta, \gamma) \in \mathbb{N}_d^s \times \mathbb{N}_d^s}$ be a complex sequence indexed by $(\beta, \gamma) \in \mathbb{N}_d^s \times \mathbb{N}_d^s$ and satisfying $y_{\beta, \gamma} = \overline{y_{\gamma, \beta}}$. Let L_y be the linear functional defined by

$$f = \sum_{(\beta, \gamma)} b_{\beta, \gamma} \mathbf{z}^\beta \bar{\mathbf{z}}^\gamma \mapsto L_y(f) = \sum_{(\beta, \gamma)} b_{\beta, \gamma} y_{\beta, \gamma}.$$

The *complex moment* matrix $\mathbf{M}_d(y)$ associated with y is the Hermitian matrix indexed by $\mathbb{N}_d^s$ such that

$$[\mathbf{M}_d(y)]_{\beta, \gamma} := L_y(\mathbf{z}^\beta \bar{\mathbf{z}}^\gamma) = y_{\beta, \gamma}, \quad \forall \beta, \gamma \in \mathbb{N}_d^s.$$

Suppose that $g = \sum_{(\beta', \gamma')} g_{\beta', \gamma'} \mathbf{z}^{\beta'} \bar{\mathbf{z}}^{\gamma'}$ is a complex polynomial. The *complex localizing* matrix $\mathbf{M}_d(gy)$ associated with g and y is the Hermitian matrix indexed by $\mathbb{N}_d^s$ such that

$$[\mathbf{M}_d(gy)]_{\beta, \gamma} := L_y(g \mathbf{z}^\beta \bar{\mathbf{z}}^\gamma) = \sum_{(\beta', \gamma')} g_{\beta', \gamma'} y_{\beta + \beta', \gamma + \gamma'}, \quad \forall \beta, \gamma \in \mathbb{N}_d^s.$$

Let $d_0 := \max \{|\beta|, |\gamma| : b_{\beta, \gamma} \neq 0\}$ and $d_i := \max \{|\beta|, |\gamma| : g_{\beta, \gamma}^i \neq 0\}$ for $i \in [t]$, where $|\cdot|$ denotes the sum of entries. Let $d_{\min} := \max \{d_0, d_1, \dots, d_m\}$. For

any $d \geq d_{\min}$, the d -th (d is called the *relaxation order*) complex moment relaxation for (CPOP) is given by

$$\begin{cases} \inf_y & L_y(f) = \langle b, y \rangle_{\mathbb{R}} \\ \text{s.t.} & \mathbf{M}_d(y) \succeq 0, \\ & \mathbf{M}_{d-d_i}(g_i y) \succeq 0, \quad i \in [t], \\ & y_{\mathbf{0}, \mathbf{0}} = 1. \end{cases} \quad (\text{Mom-C})$$

(Mom-C) and its dual problem form the complex moment-HSOS hierarchy of (CPOP). For more details on this hierarchy, we refer the reader to [12, 17].

For any $(\beta, \gamma) \in \mathbb{N}_d^s \times \mathbb{N}_d^s$, we associate it with a matrix $A_{\beta, \gamma}^0 \in \mathbb{R}^{\omega_{s,d} \times \omega_{s,d}}$ defined by

$$[A_{\beta, \gamma}^0]_{\beta', \gamma'} = \begin{cases} 1, & \text{if } (\beta', \gamma') = (\beta, \gamma), \\ 0, & \text{otherwise.} \end{cases} \quad (3.1)$$

Moreover, for each $i \in [t]$, we associate any $(\beta, \gamma) \in \mathbb{N}_{d-d_i}^s \times \mathbb{N}_{d-d_i}^s$ with a matrix $A_{\beta, \gamma}^i \in \mathbb{C}^{\omega_{s,d-d_i} \times \omega_{s,d-d_i}}$ defined by

$$[A_{\beta, \gamma}^i]_{\beta', \gamma'} = \begin{cases} g_{\beta'', \gamma''}^i, & \text{if } (\beta' + \beta'', \gamma' + \gamma'') = (\beta, \gamma), \\ 0, & \text{otherwise.} \end{cases} \quad (3.2)$$

Now for each $i = 0, 1, \dots, t$, we define the $\mathbb{R}$ -linear operator $\mathcal{A}^i$ by

$$\mathcal{A}^i(H) := \left(\langle A_{\beta, \gamma}^i, H \rangle \right)_{(\beta, \gamma) \in \mathbb{N}_{d-d_i}^s \times \mathbb{N}_{d-d_i}^s}, \quad H \in \mathbb{H}^{\omega_{s,d-d_i}}.$$

For convenience let us set $g_0 := 1$. By construction, it holds

$$\mathbf{M}_{d-d_i}(g_i y) = \sum_{(\beta, \gamma) \in \mathbb{N}_{d-d_i}^s \times \mathbb{N}_{d-d_i}^s} A_{\beta, \gamma}^i y_{\beta, \gamma} = (\mathcal{A}^i)^*(y), \quad i = 0, 1, \dots, t.$$

Therefore, one can rewrite (Mom-C) as follows:

$$\begin{cases} \inf_y & \langle b, y \rangle_{\mathbb{R}} \\ \text{s.t.} & (\mathcal{A}^i)^*(y) \succeq 0, \quad i = 0, 1, \dots, t, \\ & y_{\mathbf{0}, \mathbf{0}} = 1, \end{cases} \quad (\text{Mom-C}')$$

whose dual reads as

$$\begin{cases} \sup_{\lambda, H^i} \lambda \\ \text{s.t.} \quad \sum_{i=0}^t [\mathcal{A}^i(H^i)]_{\beta, \gamma} + \delta_{(\beta, \gamma), (0, 0)} \lambda = b_{\beta, \gamma}, \quad (\beta, \gamma) \in \mathbb{N}_d^s \times \mathbb{N}_d^s, \\ H^i \succeq 0, \quad i = 0, 1, \dots, t. \end{cases} \quad (\text{HSOS-C})$$

Note that we have used the Kronecker delta $\delta_{(\beta, \gamma), (0, 0)}$ in (HSOS-C).

Let us from now on fix any order “ $<$ ” on $\mathbb{N}^s$.

Proposition 3.1 (HSOS-C) is equivalent to the following complex SDP:

$$\begin{cases} \sup_{\lambda, H^i} \lambda \\ \text{s.t.} \quad \sum_{i=0}^t [\mathcal{A}^i(H^i)]_{\beta, \gamma} + \delta_{(\beta, \gamma), (0, 0)} \lambda = b_{\beta, \gamma}, \\ \beta \leq \gamma, \quad (\beta, \gamma) \in \mathbb{N}_d^s \times \mathbb{N}_d^s, \\ H^i \succeq 0, \quad i = 0, 1, \dots, t. \end{cases} \quad (\text{HSOS-C}')$$

Proof It suffices to show that for $\beta < \gamma$, $\sum_{i=0}^t [\mathcal{A}^i(H^i)]_{\gamma, \beta} = b_{\gamma, \beta}$ is equivalent to $\sum_{i=0}^t [\mathcal{A}^i(H^i)]_{\beta, \gamma} = b_{\beta, \gamma}$. Indeed, this equivalence follows from $b_{\gamma, \beta} = \overline{b_{\beta, \gamma}}$ and

$$\begin{aligned} \sum_{i=0}^t [\mathcal{A}^i(H^i)]_{\gamma, \beta} &= \sum_{i=0}^t \langle A_{\gamma, \beta}^i, H^i \rangle \\ &= [H^0]_{\gamma, \beta} + \sum_{i=1}^t \sum_{\substack{(\gamma', \beta') \in \mathbb{N}_{d-d_i}^s \times \mathbb{N}_{d-d_i}^s \\ (\gamma'', \beta'') \in \text{supp}(g) \\ (\gamma' + \gamma'', \beta' + \beta'') = (\gamma, \beta)}} g_{\gamma'', \beta''}^i [H^i]_{\gamma', \beta'} \\ &= \sum_{i=0}^t \langle \overline{A_{\beta, \gamma}^i}, H^i \rangle = \sum_{i=0}^t \overline{[\mathcal{A}^i(H^i)]_{\beta, \gamma}}. \end{aligned}$$

□

Let $\mathcal{A}_R^i$ and $\mathcal{A}_I^i$ be the real linear operators associated with $\mathcal{A}^i$ which are defined in a similar way as (2.2) and (2.3). With $H^i = H_R^i + H_I^i \mathbf{i}$, $b = b_R + b_I \mathbf{i}$, (HSOS-C') is equivalent to the following real SDP by using the equivalent condition (1.1):

$$\left\{ \begin{array}{l} \sup_{\lambda, Y^i} \lambda \\ \text{s.t.} \quad \sum_{i=0}^t ([\mathcal{A}_R^i(H_R^i)]_{\beta, \gamma} + [\mathcal{A}_I^i(H_I^i)]_{\beta, \gamma}) + \delta_{(\beta, \gamma), (0, 0)} \lambda = [b_R]_{\beta, \gamma}, \\ \quad \sum_{i=0}^t ([\mathcal{A}_R^i(H_R^i)]_{\beta, \gamma} - [\mathcal{A}_I^i(H_I^i)]_{\beta, \gamma}) = [b_I]_{\beta, \gamma}, \\ \quad \beta \leq \gamma, \quad (\beta, \gamma) \in \mathbb{N}_d^s \times \mathbb{N}_d^s, \\ \quad Y^i = \begin{bmatrix} H_R^i & -H_I^i \\ H_I^i & H_R^i \end{bmatrix} \succeq 0, \quad i = 0, 1, \dots, t. \end{array} \right. \quad (\text{HSOS-}\mathbb{R})$$

On the other hand, by invoking Theorem 2.2, we obtain another equivalent real SDP conversion of (HSOS-C'):

$$\left\{ \begin{array}{l} \sup_{\lambda, X^i} \lambda \\ \text{s.t.} \quad \sum_{i=0}^t \left([\mathcal{A}_R^i(X_1^i + X_2^i)]_{\beta, \gamma} + [\mathcal{A}_I^i(X_3^i - (X_3^i)^\top)]_{\beta, \gamma} \right) + \delta_{(\beta, \gamma), (0, 0)} \lambda = [b_R]_{\beta, \gamma}, \\ \quad \sum_{i=0}^t \left([\mathcal{A}_R^i(X_3^i - (X_3^i)^\top)]_{\beta, \gamma} - [\mathcal{A}_I^i(X_1^i + X_2^i)]_{\beta, \gamma} \right) = [b_I]_{\beta, \gamma}, \\ \quad \beta \leq \gamma, \quad (\beta, \gamma) \in \mathbb{N}_d^s \times \mathbb{N}_d^s, \\ \quad X^i = \begin{bmatrix} X_1^i & (X_3^i)^\top \\ X_3^i & X_2^i \end{bmatrix} \succeq 0, \quad i = 0, 1, \dots, t. \end{array} \right. \quad (3.3)$$

Proposition 3.2 (3.3) is equivalent to the following real SDP:

$$\left\{ \begin{array}{l} \sup_{\lambda, X^i} \lambda \\ \text{s.t.} \quad \sum_{i=0}^t \left([\mathcal{A}_R^i(X_1^i + X_2^i)]_{\beta, \gamma} + [\mathcal{A}_I^i(X_3^i - (X_3^i)^\top)]_{\beta, \gamma} \right) + \delta_{(\beta, \gamma), (0, 0)} \lambda = [b_R]_{\beta, \gamma}, \\ \quad \sum_{i=0}^t \left([\mathcal{A}_R^i(X_3^i - (X_3^i)^\top)]_{\beta, \gamma} - [\mathcal{A}_I^i(X_1^i + X_2^i)]_{\beta, \gamma} \right) = [b_I]_{\beta, \gamma}, \quad \beta \neq \gamma, \\ \quad \beta \leq \gamma, \quad (\beta, \gamma) \in \mathbb{N}_d^s \times \mathbb{N}_d^s, \\ \quad X^i = \begin{bmatrix} X_1^i & (X_3^i)^\top \\ X_3^i & X_2^i \end{bmatrix} \succeq 0, \quad i = 0, 1, \dots, t. \end{array} \right. \quad (\text{HSOS-}\mathbb{R}')$$

Proof We need to show that the following constraints

$$\sum_{i=0}^t \left([\mathcal{A}_R^i(X_3^i - (X_3^i)^\top)]_{\beta, \beta} - [\mathcal{A}_I^i(X_1^i + X_2^i)]_{\beta, \beta} \right) = [b_I]_{\beta, \beta} = 0, \quad \beta \in \mathbb{N}_d^s \quad (3.4)$$

in (3.3) are redundant. For each $i = 0, 1, \dots, t$ and $\beta \in \mathbb{N}_d^s$, we have

$$\begin{aligned}
 & \left\langle (A_{\beta, \beta}^i)_R, X_3^i - (X_3^i)^\top \right\rangle \\
 &= \sum_{\substack{(\beta', \gamma') \in \mathbb{N}_{d-d_i}^s \times \mathbb{N}_{d-d_i}^s \\ (\beta'', \gamma'') \in \text{supp}(g) \\ (\beta' + \beta'', \gamma' + \gamma'') = (\beta, \beta)}} \mathcal{R}(g_{\beta'', \gamma''}^i) \left([X_3^i]_{\beta', \gamma'} - [(X_3^i)^\top]_{\beta', \gamma'} \right) \\
 &= \frac{1}{2} \sum_{\substack{(\beta', \gamma') \in \mathbb{N}_{d-d_i}^s \times \mathbb{N}_{d-d_i}^s \\ (\beta'', \gamma'') \in \text{supp}(g) \\ (\beta' + \beta'', \gamma' + \gamma'') = (\beta, \beta)}} \mathcal{R}(g_{\beta'', \gamma''}^i) \left([X_3^i]_{\beta', \gamma'} + [X_3^i]_{\gamma', \beta'} - [(X_3^i)^\top]_{\beta', \gamma'} - [(X_3^i)^\top]_{\gamma', \beta'} \right) \\
 &= 0,
 \end{aligned}$$

where we have used fact that $[(X_3^i)^\top]_{\beta', \gamma'} = [X_3^i]_{\gamma', \beta'}$ and $[(X_3^i)^\top]_{\gamma', \beta'} = [X_3^i]_{\beta', \gamma'}$. It follows that $[\mathcal{A}_R^i(X_3^i - (X_3^i)^\top)]_{\beta, \beta} = \langle (A_{\beta, \beta}^i)_R, X_3^i - (X_3^i)^\top \rangle = 0$.

In addition, for each $i = 0, 1, \dots, t$ and $\beta \in \mathbb{N}_d^s$, we have

$$\begin{aligned}
 & \left\langle (A_{\beta, \beta}^i)_I, X_1^i + X_2^i \right\rangle \\
 &= \sum_{\substack{(\beta', \gamma') \in \mathbb{N}_{d-d_i}^s \times \mathbb{N}_{d-d_i}^s \\ (\beta'', \gamma'') \in \text{supp}(g) \\ (\beta' + \beta'', \gamma' + \gamma'') = (\beta, \beta)}} \mathcal{I}(g_{\beta'', \gamma''}^i) \left([X_1^i]_{\beta', \gamma'} + [X_2^i]_{\beta', \gamma'} \right) \\
 &= \frac{1}{2} \sum_{\substack{(\beta', \gamma') \in \mathbb{N}_{d-d_i}^s \times \mathbb{N}_{d-d_i}^s \\ (\beta'', \gamma'') \in \text{supp}(g) \\ (\beta' + \beta'', \gamma' + \gamma'') = (\beta, \beta)}} \mathcal{I} \left(g_{\beta'', \gamma''}^i + g_{\gamma'', \beta''}^i \right) \left([X_1^i]_{\beta', \gamma'} + [X_2^i]_{\beta', \gamma'} \right) \\
 &= 0,
 \end{aligned}$$

where we have used fact that $\mathcal{I}(g_{\beta'', \gamma''}^i + g_{\gamma'', \beta''}^i) = \mathcal{I}(g_{\beta'', \gamma''}^i + \bar{g}_{\beta'', \gamma''}^i) = 0$ and that X_1^i, X_2^i are symmetric. It follows that $[\mathcal{A}_I^i(X_1^i + X_2^i)]_{\beta, \beta} = \langle (A_{\beta, \beta}^i)_I, X_1^i + X_2^i \rangle = 0$.

Putting all above together yields (3.4). $\square$

We have proved the following theorem.

Theorem 3.3 (HSOS- $\mathbb{R}'$) is equivalent to (HSOS- $\mathbb{C}$).

Before closing the section, we compare complexity of different real SDP reformulations for complex SDP relaxations of (CPOP) in Table 1.

Table 1 Comparison of complexity of different real SDP reformulations for complex SDP relaxations of (CPOP). n_{sdp} : the maximal size of SDP matrix, m_{sdp} : the number of affine constraints

	(HSOS- $\mathbb{R}$)	(HSOS- $\mathbb{R}'$)
n_{sdp}	$2\omega_{s,d}$	$2\omega_{s,d}$
m_{sdp}	$2\omega_{s,d}^2 + 2\omega_{s,d} + \sum_{i=1}^t \omega_{s,d-d_i}$	$\omega_{s,d}^2$

4 Numerical experiments

In this section, we benchmark the performance of the two real reformulations for complex SDPs using the software TSSOS 1.2.1³ in which MOSEK 10.0 [1] is employed as an SDP solver with default settings. For comparison, we also include the results of directly solving complex SDPs obtained with Hypatia 0.8.1 [8] in Sections 4.1–4.4. We emphasize that the purpose of numerical experiments is to compare different formulations of complex SDPs and a comparison between solvers is not our focus in this paper. All numerical experiments were performed on a desktop computer with an Intel(R) Core(TM) i9-10900 CPU@2.80GHz and 64G RAM. When presenting the results, the column labelled by ‘opt’ records the optimum and the column labelled by ‘time’ records computational time in seconds. Moreover, the symbol ‘-’ means the SDP solver runs out of memory, and the symbol ‘*’ means computational time exceeds 10000 seconds.

4.1 The complex matrix completion problem

The complex matrix completion problem seeks to recover a low-rank matrix $M \in \mathbb{C}^{s \times t}$ from a subset of entries $\{M_{ij}\}_{(i,j) \in \Omega}$ [6, 7], which could be tackled via the convex optimization problem:

$$\begin{cases} \inf_{Z \in \mathbb{C}^{s \times t}} \|Z\|_* \\ \text{s.t. } Z_{ij} = M_{ij}, \quad \forall (i, j) \in \Omega, \end{cases} \quad (4.1)$$

where $\|Z\|_* := \text{Tr}(Z^H Z)^{\frac{1}{2}}$ is the nuclear norm of Z . One can equivalently cast (4.1) as a complex SDP:

$$\begin{cases} \inf_{X \in \mathbb{H}_n} \text{Tr}(X) \\ \text{s.t. } Z_{ij} = M_{ij}, \quad \forall (i, j) \in \Omega, \\ X = \begin{bmatrix} U & Z^H \\ Z & V \end{bmatrix} \succeq 0. \end{cases} \quad (4.2)$$

In this subsection, we generate random instances of the complex matrix completion problem (4.1) as follows: 1) select $\Omega \subseteq [s] \times [s]$ uniformly at random from all subsets with cardinality $10s$ for $s \in \{20, 30, 40, 50, 60, 70\}$; 2) set $M = M_1 M_2^H$ with the real

³ TSSOS is freely available at <https://github.com/wangjie212/TSSOS>.

Table 2 The complex matrix completion problem

s	(PSDP- $\mathbb{R}$)			(PSDP- $\mathbb{R}'$)			(PSDP- $\mathbb{C}$)	
	m_{sdp}	opt	time	m_{sdp}	opt	time	opt	time
20	1948	315.882	0.62	308	315.882	0.06	315.882	3.46
30	4168	456.591	5.38	508	456.591	0.11	456.591	31.1
40	7160	670.611	62.7	680	670.611	0.37	670.611	153
50	11010	861.510	175	910	861.510	0.77	861.510	626
60	15602	925.590	573	1082	925.590	1.07	925.590	1899
70	21056	1108.93	1337	1316	1108.93	1.81	1108.93	4870

and imaginary parts of entries of $M_1 \in \mathbb{C}^{s \times 5}$ and $M_2 \in \mathbb{C}^{s \times 5}$ being selected i.i.d. from the standard normal distribution. The related results are shown in Table 2. From the table, we see that the reformulation (HSOS- $\mathbb{R}'$) is $1 \sim 3$ magnitudes faster than the reformulation (HSOS- $\mathbb{R}$).

4.2 Minimizing a random complex quartic polynomial over the unit sphere

Our second example is to minimize a complex quartic polynomial over the unit sphere:

$$\begin{cases} \inf_{\mathbf{z} \in \mathbb{C}^s} & [\mathbf{z}]_2^H Q [\mathbf{z}]_2 \\ \text{s.t.} & |z_1|^2 + \cdots + |z_s|^2 = 1, \end{cases} \quad (4.3)$$

where $[\mathbf{z}]_2$ is the column vector of monomials in $\mathbf{z}$ up to degree two and $Q \in \mathbb{H}^{|\mathbf{z}|_2}$ is a random Hermitian matrix whose entries are selected with respect to the standard normal distribution.

We approach (4.3) for $s = 5, 7, \dots, 15$ with the second and third HSOS relaxations. The related results are shown in Table 3. From the table, we see that the reformulation (HSOS- $\mathbb{R}'$) is several ($2 \sim 7$) times as fast as the reformulation (HSOS- $\mathbb{R}$), and the speedup becomes more significant as the SDP size grows.

4.3 Minimizing a random complex quartic polynomial with unit-norm variables

The next example is to minimize a random complex quartic polynomial with unit-norm variables:

$$\begin{cases} \inf_{\mathbf{z} \in \mathbb{C}^s} & [\mathbf{z}]_2^H Q [\mathbf{z}]_2 \\ \text{s.t.} & |z_i|^2 = 1, \quad i = 1, \dots, s, \end{cases} \quad (4.4)$$

where $Q \in \mathbb{H}^{|\mathbf{z}|_2}$ is a random Hermitian matrix whose entries are selected with respect to the uniform probability distribution on $[0, 1]$.

We approach (4.4) for $s = 5, 7, \dots, 15$ with the second and third HSOS relaxations. The related results are shown in Table 4. From the table, we see that the reformulation

Table 3 Minimizing a random complex quartic polynomial over the unit sphere

s	d	(HSOS- $\mathbb{R}$)			(HSOS- $\mathbb{R}'$)			(HSOS- $\mathbb{C}'$)	
		m_{sdp}	opt	time	m_{sdp}	opt	time	opt	time
5	2	966	-11.2409	0.11	441	-11.2409	0.05	-11.2409	0.12
	3	6846	-9.47725	8.13	3136	-9.47725	2.00	-9.47725	6.54
7	2	2736	-14.2314	0.97	1296	-14.2314	0.28	-14.2314	0.59
	3	30372	-11.0407	389	14400	-11.0407	57.0	-11.0407	474
9	2	6270	-19.0019	5.73	3025	-19.0019	1.62	-19.0019	4.61
	3	100320	-	-	48400	-15.5614	1944	-	-
11	2	12480	-22.8630	31.7	6084	-22.8630	6.67	-22.8630	32.3
	3	271882	-	-	132496	-	-	-	-
13	2	22470	-25.6352	145	11025	-25.6352	23.5	-25.6352	174
	3	639450	-	-	313600	-	-	-	-
15	2	37536	-29.1672	585	18496	-29.1672	86.1	-29.1672	802
	3	1351976	-	-	665856	-	-	-	-

Table 4 Minimizing a random complex quartic polynomial with unit-norm variables

s	d	(HSOS- $\mathbb{R}$)			(HSOS- $\mathbb{R}'$)			(HSOS- $\mathbb{C}'$)	
		m_{sdp}	opt	time	m_{sdp}	opt	time	opt	time
5	2	734	-24.4919	0.10	271	-24.4919	0.03	-24.4919	0.21
	3	4474	-24.4919	2.34	1281	-24.4919	0.26	-24.4919	10.9
7	2	2202	-56.5289	0.65	869	-56.5289	0.16	-56.5289	1.15
	3	21158	-46.7128	132	6637	-46.7128	7.44	-46.7128	520
9	2	5242	-114.342	4.62	2161	-114.342	0.73	-114.342	5.29
	3	73312	-	-	24691	-81.2676	184	-	-
11	2	10718	-202.436	32.1	4555	-202.436	3.86	-202.436	30.0
	3	206188	-	-	73327	-	-	-	-
13	2	19686	-338.041	126	8555	-338.041	12.7	-338.041	162
	3	499438	-	-	185277	-	-	-	-
15	2	33394	-514.226	678	14761	-514.226	55.1	-514.226	705
	3	1081514	-	-	414841	-	-	-	-

(HSOS- $\mathbb{R}'$) is about one magnitude faster than the reformulation (HSOS- $\mathbb{R}$), and again the speedup becomes more significant as the SDP size grows.

Table 5 Minimizing a randomly generated sparse complex quartic polynomial over multi-spheres

l	(HSOS- $\mathbb{R}$)			(HSOS- $\mathbb{R}'$)			(HSOS- $\mathbb{C}'$)	
	m_{sdp}	opt	time	m_{sdp}	opt	time	opt	time
40	23090	-98.9240	3.12	12529	-98.9240	2.06	-98.9240	886
80	46768	-197.577	12.6	25549	-197.577	8.07	-197.577	5433
120	70958	-292.024	30.1	38871	-292.024	19.0	*	*
160	94278	-389.652	45.9	51563	-389.652	30.7	*	*
200	117526	-482.684	84.5	64185	-482.684	37.7	*	*
240	140298	-578.896	130	76389	-578.896	59.5	*	*
280	162504	-671.047	173	89241	-671.047	65.4	*	*
320	187528	-766.403	206	102171	-766.403	88.5	*	*
360	210370	-866.771	291	114589	-866.771	147	*	*
400	233396	-963.137	297	127173	-963.137	138	*	*

4.4 Minimizing a randomly generated sparse complex quartic polynomial over multi-spheres

Given $l \in \mathbb{N} \setminus \{0\}$, we randomly generate a sparse complex quartic polynomial as follows: Let $f = \sum_{i=1}^l f_i \in \mathbb{C}[z_1, \dots, z_{5(l+1)}, \bar{z}_1, \dots, \bar{z}_{5(l+1)}]$,⁴ where for all $i \in [l]$, $f_i = \bar{f}_i \in \mathbb{C}[z_{5(i-1)+1}, \dots, z_{5(i-1)+10}, \bar{z}_{5(i-1)+1}, \dots, \bar{z}_{5(i-1)+10}]$ is a sparse complex quartic polynomial whose coefficients (real/imaginary parts) are selected with respect to the uniform probability distribution on $[-1, 1]$. Then we consider the following CPOP:

$$\begin{cases} \inf_{\mathbf{z} \in \mathbb{C}^{5(l+1)}} f(\mathbf{z}, \bar{\mathbf{z}}) \\ \text{s.t.} \quad \sum_{j=1}^{10} |z_{5(i-1)+j}|^2 = 1, \quad i = 1, \dots, l. \end{cases} \quad (4.5)$$

The sparsity in (4.5) can be exploited to derive a sparsity-adapted complex moment-HSOS hierarchy [17]. We solve the second sparse HSOS relaxation of (4.5) for $l = 40, 80, \dots, 400$. The results are displayed in Table 5. From the table we see that the reformulation (HSOS- $\mathbb{R}'$) is $1.5 \sim 2$ times as fast as the reformulation (HSOS- $\mathbb{R}$). Moreover, for this problem, solving the original complex SDP with Hypatia is extremely slow probably due to the fact that the SDPs contain many PSD blocks.

4.5 Application to the AC-OPF problem

The AC-OPF is a central problem in power systems, which aims to minimize the generation cost of an alternating current transmission network under physical and operational constraints. Mathematically, it can be formulated as the following CPOP:

⁴ $\mathbb{C}[\mathbf{z}, \bar{\mathbf{z}}]$ denotes the ring of complex polynomials in variables $\mathbf{z}, \bar{\mathbf{z}}$.

Table 6 The results for the AC-OPF problem. s : the number of CPOP variables; t : the number of CPOP constraints

Case	s	t	(HSOS- $\mathbb{R}$)			(HSOS- $\mathbb{R}'$)		
			m_{sdp}	opt	time	m_{sdp}	opt	time
14_ieee	19	147	2346	1.9940e3	0.19	422	1.9940e3	0.10
30_ieee	36	297	4828	8.1959e3	0.73	836	8.1960e3	0.37
30_as	36	297	4828	5.0371e2	0.55	836	5.0371e2	0.24
39_epri	49	361	5270	1.3568e5	0.74	966	1.3579e5	0.54
89_pegase	101	1221	57888	9.4098e4	63.6	10262	9.4101e4	15.1
57_ieee	64	563	11102	3.6644e4	2.36	2008	3.6644e4	1.06
118_ieee	172	1325	25374	9.3216e4	8.27	4471	9.3216e4	2.68
162_ieee_dtc	174	1809	64874	1.0492e5	43.4	11327	1.0495e5	13.8
179_goc	208	1827	25712	6.0859e5	10.3	4368	6.0860e5	3.57
240_pserc	383	3039	52172	2.8153e6	31.9	9243	2.8170e6	10.7
300_ieee	369	2983	53946	5.3037e5	40.6	9647	5.3037e5	10.6
500_goc	671	5255	90502	3.9697e5	89.8	15918	3.9697e5	25.4
588_sdet	683	5287	79362	1.9799e5	91.7	13933	1.9749e5	21.3
793_goc	890	7019	104978	1.1194e5	105	18536	1.1222e5	31.5
1888_rte	2178	18257	280580	1.2537e6	939	47205	1.2545e6	180
2000_goc	2238	23009	455530	9.1876e5	2087	77974	9.1881e5	439

$$\begin{cases} \inf_{V_i, S_k^g} & \sum_{k \in G} (\mathbf{c}_{2k}(\mathcal{R}(S_k^g))^2 + \mathbf{c}_{1k}\mathcal{R}(S_k^g) + \mathbf{c}_{0k}) \\ \text{s.t.} & \angle V_r = 0, \\ & \mathbf{S}_k^{gl} \leq S_k^g \leq \mathbf{S}_k^{gu}, \quad \forall k \in G, \\ & \mathbf{v}_i^l \leq |V_i| \leq \mathbf{v}_i^u, \quad \forall i \in N, \\ & \sum_{k \in G_i} S_k^g - \mathbf{S}_i^d - \mathbf{Y}_i^{sh} |V_i|^2 = \sum_{(i,j) \in E_i \cup E_i^R} S_{ij}, \quad \forall i \in N, \\ & S_{ij} = (\bar{\mathbf{Y}}_{ij} - \mathbf{i} \frac{\mathbf{b}_{ij}^c}{2}) \frac{|V_i|^2}{|\mathbf{T}_{ij}|^2} - \bar{\mathbf{Y}}_{ij} \frac{V_i \bar{V}_j}{\mathbf{T}_{ij}}, \quad \forall (i, j) \in E, \\ & S_{ji} = (\bar{\mathbf{Y}}_{ij} - \mathbf{i} \frac{\mathbf{b}_{ij}^c}{2}) |V_j|^2 - \bar{\mathbf{Y}}_{ij} \frac{\bar{V}_i V_j}{\mathbf{T}_{ij}}, \quad \forall (i, j) \in E, \\ & |S_{ij}| \leq \mathbf{s}_{ij}^u, \quad \forall (i, j) \in E \cup E^R, \\ & \theta_{ij}^{\Delta l} \leq \angle(V_i \bar{V}_j) \leq \theta_{ij}^{\Delta u}, \quad \forall (i, j) \in E, \end{cases} \quad (4.6)$$

where V_i is the voltage, S_k^g is the power generation, S_{ij} is the power flow (all are complex variables; $\angle \cdot$ stands for the angle of a complex number) and all symbols in boldface are constants. Notice that G is the collection of generators and N is the collection of buses. For a full description on the AC-OPF problem, we refer the reader to [3] as well as [5].

We select test cases from the AC-OPF library **PGLiB-OPF** [3]. For each case, we solve the minimal relaxation step of the sparse HSOS hierarchy [17]. The results are displayed in Table 6. From the table, we again see that the reformulation (HSOS- $\mathbb{R}'$) is several (1.4 ~ 5) times as fast as the reformulation (HSOS- $\mathbb{R}$).

Summary of the benchmark. The benchmark confirms our theoretical findings and demonstrates that our new real reformulation of complex SDPs is indeed more efficient than the classical real reformulation. Especially, the speedup factor gets even bigger as the SDP size grows and may reach several magnitudes on certain instances. It should be also noted that solving the original complex SDP with *Hypatia* is even slower than solving the classical real reformulation with MOSEK, probably because MOSEK is commercial and the implementations of SDP algorithms based on real numbers are more mature and robust.

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Declarations

Conflicts of Interest The authors declare that they have no conflict of interest.

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